A Multi-Agent AI System for Total Cost Analysis

1. Problem & Motivation

Consumers face many choices: online stores, in-store pickup, or physical shopping. The true cost of a product includes price, shipping, membership benefits, and time spent obtaining it. Comparing all these manually is slow and error-prone.

Target Users: Anyone who wants to find the best option considering total cost and convenience, including those who prioritize lowest price, speed, or membership perks (e.g., Amazon Prime, Walmart+).

Why an Agent System: The task involves multiple steps:

- Identifying the exact product
- Fetching data from online and local stores
- Calculating the final cost with shipping and membership rules
- Presenting a clear comparison

A system of specialized agents can handle each step efficiently, adapt to new retailers, and remain transparent.

2. Use Cases

- **Full Cost Comparison:** User searches for a coffee maker. Agents fetch online and in-store prices, calculate delivery or travel time, and produce a table comparing total costs.

- **General Product Search:** User wants an extension cord, any brand. Agents fetch prices from multiple stores and rank by total cost, including delivery or travel time.
- **Exact Product Search:** User wants a specific moisturizer. Agents compare prices of that exact product across online and local stores, applying membership perks and delivery calculations.
- **Prioritizing Speed:** User needs a phone charger fast. Agents find same-day delivery and local pickup options, calculate times, and rank options by speed and cost.
- **Membership Evaluation:** User with Walmart+ shops for groceries. Agents compare delivery vs. in-store totals, applying membership benefits.

3. System Design

Agents & Roles:

- **Query Interpreter:** Standardizes product requests
- **Online Price Fetchers:** Gather live e-commerce prices
- **Local Store Agent:** Estimates nearby store prices
- **Cost Calculator:** Adds shipping, membership perks, and location info
- **Data Validator:** Ensures data accuracy
- **Comparison Analyst:** Ranks options by cost, speed, or balance

Tools & Data: APIs where available, simulated data for missing stores, rules-based shipping/membership logic, interactive Streamlit interface.

4. Feasibility & Risks

- **Challenges:** Real-time store prices, API limits, shipping complexity
- **Mitigations:** Use estimates, cached data, simplified shipping rules
- **Constraints:** User privacy is protected, system runs on standard computers

5. Timeline

1. Proposal & design
2. Front-end development
3. Multi-agent workflow integration
4. Testing and pilot scenarios
5. Final system, reporting, and presentation

6. Conclusion

The system provides a clear, easy-to-use way for consumers to compare total costs across channels. Specialized agents gather data, apply rules, and produce actionable comparisons, demonstrating a functional multi-agent workflow within the project scope.